MACHINE LEARNING PROJECT

Diabetes Prediction Using ANN

From health measurements to an explainable prediction

Artificial neural network • Model comparison
Calibrated ensemble • Local prediction app

768
Historical records

8
Numeric input features

5
Models in the ensemble

Presented by: [Your name] | [Register number] | [Department / Institution]

Problem statement and objectives

Can an ANN classify diabetes outcomes from health measurements and explain its predictions in a usable application?

01 Build

Clean the data and train an ANN with a reproducible pipeline.

02 Compare

Evaluate four alternative learners and a calibrated ensemble.

03 Explain

Show feature contributions, model disagreement and what-if results.

Scope: academic outcome classification using a historical dataset.

Source: docs/project_brief.md; docs/methodology.md

Literature Survey and Research Gap

Five diabetes studies and five papers on the methods used in this project

Paper / author and year	Approach	Connection to this project
Sisodia & Sisodia (2018) — Diabetes classification algorithms	DT, SVM and Naive Bayes	Compare multiple baseline learners
Zou et al. (2018) — Diabetes with machine learning	RF, DT, neural nets; PCA / mRMR	Test feature selection and ANN
Sneha & Gangil (2019) — Diabetes and optimal feature selection	Feature selection; classifier comparison	Measure feature-selection benefit
Tasin et al. (2023) — Diabetes prediction with explainable AI	ML ensembles, SHAP / LIME, app	Explain the deployed classifier
Hasan et al. (2025) — Transparent diabetes: AutoML + XAI	AutoGluon, XAI, Streamlit interface	Compare models and what-if outputs
Guo et al. (2017) — Calibration of modern neural networks	Confidence calibration; temperature scaling	Measure Brier and ECE separately
Lundberg & Lee (2017) — Unified interpretation of predictions	SHAP additive feature attribution	Explain full pipeline; check additivity
Breiman (2001) — Random Forests	Randomized ensembles of decision trees	Tree-ensemble comparison baseline
Friedman (2001) — A gradient boosting machine	Stagewise additive / gradient boosting	Boosting comparison baseline
Cortes & Vapnik (1995) — Support-vector networks	Margin-based kernel classification	Nonlinear SVM comparison baseline

Gap tested: do selection, voting and calibration improve a fixed holdout? Evaluate trade-offs and explain the full pipeline.

Sources: linked paper titles; full references in speaker notes and docs/literature_survey_10_papers.md

Dataset and preprocessing

Eight input features

Pregnancies • Glucose • Blood pressure

Skin thickness • Insulin • BMI

Diabetes pedigree score • Age

Target: Outcome = 0 or 1

Class distribution



500 negative | 268 positive

Zero handling → Median imputation

Missingness indicators → Standard scaling

Training-CV selection → 5 predictors

Stratified split: 460 train / 154 validation / 154 test • Seed 42

Source: docs/dataset_card.md; reports/results.json

CORE MODEL

ANN architecture and training

5 selected inputs

After preprocessing

→

32 hidden units

ReLU activation

→

1 output

Sigmoid probability

Selected configuration

Adam optimizer • Learning rate 0.003

L2 regularization 0.001 • Early stopping

Training and tuning

5-fold stratified CV • 6 candidates/model

Best ANN training-CV ROC-AUC: 0.821

Source: reports/tuning_summary.json; docs/methodology.md

Prediction framework and application

ANN + Logistic Regression + Random Forest + Gradient Boosting + SVM

Equal-weight voting → **Sigmoid calibration** → **Threshold = 0.33**

Prediction interface

Enter 8 features → Probability and class

Missing-value and extrapolation notes

Interpretation tools

Exact Shapley feature contributions

Component disagreement • What-if curves

Source: docs/methodology.md; src/diabetes/predict.py; web/

Measured results on the held-out test set

Test ROC-AUC by model



Deployed ensemble

Validation threshold: 0.33

71.4% Accuracy	**70.4%** Recall
0.633 F1 score	**72.0%** Specificity

AUC 95% interval: 0.736–0.876

Gradient boosting achieved the highest test AUC; the ensemble was the prespecified deployed framework.

Source: reports/test_metrics_*.csv; reports/bootstrap_intervals.json

Conclusion and future work

Completed

ANN training and model comparison

Prediction app with explanations

Key finding

Useful predictive ranking, with trade-offs

Calibration worsened the Brier score

Limitations

Historical cohort; missing measurements

No external validation; non-causal explanations

Next steps

Test on diverse independent data

Assess fairness, calibration and thresholds

Academic demonstrator • Outcome classification • Not validated for clinical diagnosis

Source: docs/experiment_notes.md; reports/results.json